Eric Zhang

Email: ezhang1218@gmail.com | Personal Website: https://ezhang1218.github.io | Phone: 9148743315

Education

University of North Carolina at Chapel Hill — Chapel Hill, NC
PhD Candidate, Department of Biostatistics — *Aug. 2021 – Dec 2026*
Research Areas: High-dimensional data, Dimension Reduction, Statistical Modeling, Machine Learning.

Cornell University, BA — Ithaca, NY
Double Major: Mathematics, Statistics. Minor: Computer Science. GPA: 3.70 — *Aug. 2017 – May 2021*

Publications

Zhang, E., Li, D. (2025). Contrastive Functional Principal Components Analysis. *Proceedings of the AAAI Conference on Artificial Intelligence, 39(21).* [Link] [Code]

Zhang, E., Love, M., Li, D. (2025). Contrastive CUR: Interpretable Joint Feature and Sample Selection for Case-Control Studies. *arXiv:2508.11557.* [Link] [Code]

Hawke, S.*, **Zhang, E.***, Chen, J.*, Li, D. (2025). Contrastive Dimension Reduction: A Systematic Review. *Wiley Interdisciplinary Reviews: Computational Statistics18, no.2: e70065.* [Link] [Code]

Muhlebach, M., **Zhang, E.**, et al. (2025). Association Between Inhaled Antibiotic Use and Treatment-Emergent Organisms among Canadian People with CF. *Journal of Cystic Fibrosis.*

Muhlebach, M., **Zhang, E.**, et al. (2024). Changes in factors associated with inhaled antibiotic prescriptions for people with cystic fibrosis over time in the U.S. *Journal of Cystic Fibrosis, 24(1), 98–104.*

Experience

JPMorgan Chase — Jun. 2025 – Aug. 2025
Quantitative Analytics Summer Associate — *New York*

- Applied XGBoost and evaluated survival model approaches on various vehicle lease exits outcome. XGBoost significantly increased AUC, and the survival model achieved a comparable Harrell's C-index.
- Examined time-varying features in relation to time to default outcomes using Cox proportional hazards models with counting process and generated time-dependent AUCs.

UNC at Chapel Hill — Aug. 2021 – Present
Graduate Research Assistant, PhD Candidate — *North Carolina*

Statistical Consulting

- Developed longitudinal causal analysis pipelines using GEE and IPTW-based survival models to evaluate time-varying inhaled antibiotic exposure and recurrent respiratory organism acquisition in cystic fibrosis patients.
- Implemented year-specific propensity score models and Andersen–Gill counting process survival models for recurrent events.

LLM Projects

- Artifact detection in ECG data is a complex problem. Signal-to-noise ratio is small, and even physicians have difficulty identifying them. We look to improve detection by fine-tuning multimodal large language models (LLMs) such as Qwen and LLaVA on ECG image data to detect artifacts. Additionally, we utilize CLIP embeddings and trained downstream classifiers, achieving an AUC of 0.95.
- Implemented a light-weight foundation model for neuroimaging data analysis, taking advantage of both LLM and existing brain knowledge. Embeddings are subsequently used to do downstream tasks such as disease classification and cognitive trait prediction.
- Experience building a LLM fine-tuning pipeline using LoRA and HuggingFace Transformers, experimenting with multiple instruction datasets and open-source language models on GPU-based HPC infrastructure.

Software and ML Engineering

- Helped develop and maintain Lola, an AI-powered food recommendation system that scrapes product pages and extracts nutritional data using OCR and LLM pipelines (AWS Bedrock / Claude).

- Implemented embedding-based similarity search using vector databases and LLM embeddings to recommend healthier alternatives.

Ernst & Young Jul. 2020 – Aug. 2020
Quantitative Advisory Intern *New York*

- Performed fair value (MTM) valuation for various financial products such as CDS, FX forwards, option, etc.

Awards

National Institude of Environmental Health Sciences
T32 Training Grant in Environmental Biostatistics.

National Chess Master
United States Chess Federation Certified.

Skills

Languages : Python, R, SQL, Pytorch, SAS.

Machine Learning: Andrew Ng's Stanford courses: Machine Learning, Neural Networks & Deep Learning, Hyperparameter tuning, Regularization and Optimization.

Data Science: Causal Inference, Statistical modeling, Hypothesis testing, Python (e.g.scikit-learn, numpy, pandas, matplotlib), A/B Testing